EXACT FIBER ELIMINATION FOR THE LEVI–CIVITA CONVEXITY GATES OF THE PLANAR RESTRICTED THREE-BODY PROBLEM, WITH A GLOBAL CONVEXITY RECONNAISSANCE

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Abstract. Computer-assisted strict-convexity certification of the regularized energy surfaces of the planar circular restricted three-body problem (PCR3BP) reduces to the simultaneous positivity of two scalar *gates*, T and D, over a three-dimensional domain: a two-dimensional Levi–Civita base together with a fiber angle $\phi \in \mathbb{S}^1$. We prove four exact algebraic identities giving an exact tangential fiber minimum and a rigorous determinant lower bound without a fiber sweep. The tangential gate satisfies $\min_\phi T = \mathcal{T}$ with $\mathcal{T}$ an explicit function of the base alone, attained at a computable angle; the curvature-determinant gate admits an exact five-term harmonic decomposition whose second-harmonic amplitude is $128F^2s$ for the structure matrices at hand, yielding the rigorous lower bound $D \geq \mathcal{D} := A_0 - \sqrt{A_1^2 + B_1^2} - 128F^2s$, which is sharp on the collar. Thus the tangential condition is reduced exactly from three variables to two, while the determinant condition is replaced by a sufficient two-variable condition. All identities are confirmed by an exact symbolic audit whose residuals vanish identically. As a numerical spot check, pointwise evaluations of the reduced gates are reported near the Earth–Moon mass ratio (the regularized component being the Moon's); this is not an interval certificate. Using the reduced gates we compute a global reconnaissance of Levi–Civita convexity across the mass–energy plane, parametrized by the mass fraction m_r of the regularized primary. In the numerical scans near the critical energy, the apparent non-convex window is large and asymmetric: at $\Delta c = 10^{-4}$ the sampled onset lies between $m_r = 0.04$ and 0.05, and negative sampled values persist through the largest sampled fraction $m_r = 0.999$, so the scans suggest LC-convexity there only for components regularized at a sufficiently light primary. The computed heavy-side non-convexity persists at the sampled masses approaching the rotating-Kepler limit; it does not extend an Euler-model theorem. We record a numerical observation that the non-convexity region avoids the coordinate axes in the sampled comparable-mass cases and forms lateral lobes on which the sampled curvature failure has one negative direction, the tangential gate staying positive, while the base remains radially star-shaped; and we discuss the limits of inferring dynamics from embedding convexity. A refinement of the reconnaissance down to energy offsets of 10^{-7} suggests that even the Moon component of the Earth–Moon system ($m_r \approx 0.0122$) loses LC-convexity on an ultrathin shell of width $\approx 9.5 \times 10^{-6}$ above the critical energy, where the Lagrange point enters the surface; for the Earth component ($m_r \approx 0.9878$), the scans suggest a macroscopic near-critical non-convexity window and positive sampled gates at offsets $\Delta c \gtrsim 1.2 \times 10^{-2}$. This numerically locates a possible limitation of the LC route near the critical limit.

1. Introduction

1.1. **Background.** Let $\mu \in (0, 1)$ be the mass ratio of the two primaries of the planar circular restricted three-body problem (PCR3BP). Below the first critical energy, the regularized bounded components are contact manifolds diffeomorphic to $\mathbb{R}P^3$, with Levi–Civita double covers diffeomorphic to $\mathbb{S}^3$ [1]. A central structural question, going back to Birkhoff [2], is whether the retrograde periodic orbit bounds a disk-like global surface of section, interpreted rationally on the quotient. Hofer, Wysocki and Zehnder [3] establish the existence of a disk-like global section for a closed

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strictly convex hypersurface in $\mathbb{R}^4$. On a lens space, Hryniewicz and Salomão [4] obtain rational open books from dynamical convexity when the proposed binding is a p-unknot with rational self-linking number $-1/p$. Identifying a prescribed orbit as the binding requires these additional topological conditions; it does not follow from a local convexity test alone.

The standard double cover is the Levi–Civita (LC) regularization [6], which represents the bounded component by a compact hypersurface $\Sigma_{\mu,c} = \{K_c = 0\} \subset \mathbb{R}^4 \cong \mathbb{C}^2$. Convexity depends on the chosen embedding. Kim [5] studies the mechanical Euler problem of two fixed centers: his Theorem 1.1 concerns an elliptic-coordinate embedding, whereas his non-convexity result concerns its standard LC image. Remark 1.2 in the cited arXiv version discusses small rotation perturbations. These statements do not provide a mass threshold for the full rotating PCR3BP; the model distinction is discussed in Section 5.

Convexity tests for Hamiltonian energy levels were developed earlier by Salomão [7], for mechanical systems including non-regular levels. In celestial mechanics, global sections in suitable lunar parameter ranges were constructed by Albers, Fish, Frauenfelder, Hofer and van Koert [8]; dynamical convexity and convex embeddings of the rotating Kepler problem were studied in [9, 10]. Rational global sections with prescribed bindings also arise in [11, 4]. More recently, Liu and Salomão [12] construct finite energy foliations near the first critical level and prove Birkhoff's retrograde-orbit conclusion for mass ratios sufficiently close to $1/2$ at all subcritical energies; their Hill-problem work [13] establishes the corresponding subcritical conclusion and near-critical foliations in that model. These results motivate approaches beyond convexity of one fixed LC embedding. The present contribution remains the algebraic elimination of the fiber variable and a separately identified numerical reconnaissance.

1.2. **The computational bottleneck.** A computer-assisted convexity check on a fixed parameter box can proceed by verifying, with interval arithmetic, that a pair of scalar *gates* T and D are strictly positive at every point of the regularized surface. After the LC reduction the surface fibers over a compact two-dimensional base $z = (z_1, z_2)$ by an angle $\phi \in \mathbb{S}^1$, so each gate is a function of the three real variables (z_1, z_2, ϕ) (with the parameters μ, c fixed). The angular variable multiplies the cost of certification: every base cell must be checked against the whole circle of fiber directions, and the interval enclosures of trigonometric expressions inflate the bounds. Eliminating ϕ analytically gives an exact two-variable minimum for T and a sufficient two-variable lower bound for D, removing the fiber sweep when these reduced positivity conditions hold.

1.3. **Contributions.** This paper makes the following contributions.

(i) **Exact fiber elimination.** We prove four algebraic identities (Lemmas 3.1–3.5) that remove the fiber variable from the tangential test and yield a sufficient determinant test on the two base variables z. For the tangential gate the reduction is an exact minimum, $\min_\phi T = \mathcal{T}(z)$, attained at an explicit angle. For the curvature-determinant gate we obtain an exact harmonic decomposition and a rigorous, collar-sharp lower bound $D \geq \mathcal{D}(z)$. Section 3.

(ii) **Symbolic certification.** Every identity is confirmed by an exact computer-algebra audit whose eight residuals vanish identically; the audit is short, deterministic, and included in the reproducibility package (Appendix A).

(iii) **Application.** Pointwise evaluations of the reduced gates are reported near the Earth–Moon mass ratio as a reproducible numerical spot check (Section 4).

(iv) **Global reconnaissance.** Using the reduced gates we compute a floating-point map of LC-convexity across the mass–energy plane, in the mass fraction m_r of the regularized primary. At the sampled offset $\Delta c = 10^{-4}$, the sampled onset lies between $m_r = 0.04$ and 0.05, and negative sampled values persist through $m_r = 0.999$, the largest sampled fraction; no negative determinant value is found for $\Delta c \gtrsim 0.03$ at the mass ratios scanned (Section 5).

(v) **A dynamical-relevance observation.** We record numerical evidence that the non-convexity region avoids the coordinate axes in the sampled comparable-mass cases and that, on the lateral lobes where it lives, the sampled failure has one negative direction on a base that stays radially star-shaped; and we discuss the remaining limitations of this embedding-based test (Sections 6–7).

Contributions (i)–(ii) are rigorous; (iii) is a finite numerical spot check, not an interval certification; (iv)–(v) are floating-point reconnaissance and are labeled as such throughout.

No fixed parameter pair or parameter region receives a rigorous numerical convexity certificate in this work. The rigorous results are the exact identities and sufficient criterion of Section 3, with the symbolic audits of Appendix A; the computations in Sections 4–6 are finite floating-point samples. In particular, the parameter rectangle in Section 4, the mass and energy thresholds in Section 5, and the sampled lobe geometry in Section 6 have not been interval-certified. No interval-certification package is claimed.

2. The regularized surface and the convexity gates

2.1. **Levi–Civita coordinates.** In rotating coordinates centered at the primary being regularized, with the other primary at $(1,0)$, pass to Levi–Civita coordinates $z=(z_1,z_2)\in\mathbb{R}^2$ via $q=2\bar{z}^2$. Writing $s=|z|^2=z_1^2+z_2^2$ and $a=z_1^2-z_2^2$, the regularized defining function for the bounded component takes the form

$$F(z)=1-\mu+\frac{2\mu s}{\sqrt{1-4a+4s^2}}-2cs+\mu^2 s+4s^3-4\mu s a, \tag{1}$$

where c is one half of the Jacobi constant. Let $B_{\mu,c}\subset\mathbb{R}^2$ denote the relevant closed bounded base component on which $F\geq 0$. Its boundary $\partial B_{\mu,c}$, given by $F=0$, is the locus called the *collar* in this paper. This one-dimensional boundary must not be confused with the three-dimensional regularized hypersurface $\Sigma_{\mu,c}=\{K_c=0\}\subset\mathbb{R}^4$ introduced in Section 1. The hypersurface projects onto $B_{\mu,c}$; the fiber is a circle parametrized by ϕ when $F>0$ and collapses at the collar. Note that F depends on z only through the pair (s,a); this is used repeatedly below.

Remark 2.1 (Mass convention)**.** In (1) the parameter μ is the mass fraction of the *distant* primary, located at $q=(1,0)$; the *regularized* primary, at the origin, has mass fraction $m_r=1-\mu$. Indeed, writing $\Omega(q)=\frac{1}{2}|q-b|^2+(1-\mu)/|q|+\mu/|q-e|$ for the effective potential of the pair with mass $1-\mu$ at the origin, mass μ at $e=(1,0)$ and barycentre $b=(\mu,0)$, the substitution $q=2\bar{z}^2$ gives the exact identity $F=|q|\,(\Omega(q)-c)$, reproducing (1) term by term ($\sqrt{1-4a+4s^2}=|q-e|$, $\mu^2 s=\frac{1}{2}\mu^2|q|$ from the barycentric offset, $-4\mu s a=-\mu q_1|q|$); the identity is verified symbolically in the reproducibility package.

2.2. **The two gates.** To specify the coordinates and normalization, let $q=(q_1,q_2)$ be primary-centered rotating coordinates, let $e=(1,0)$, and let p be the canonical momentum. The Hamiltonian at energy $-c$ is

$$H(q,p)=\frac{1}{2}|p|^2+q_2p_1-(q_1-\mu)p_2-\frac{1-\mu}{|q|}-\frac{\mu}{|q-e|}.$$

Write $R(z)=\left(\begin{smallmatrix} z_1 & -z_2\\ -z_2 & -z_1\end{smallmatrix}\right)$, so that the map $q=2\bar{z}^2$ has derivative $Dq=4R(z)$. Its canonical cotangent lift is $w=(Dq)^{\top}p$. Set $v=w/4$ and

$$\beta(z)=\big(-(2s+\mu)z_2,\ (2s-\mu)z_1\big),\qquad K_c(z,v)=|v+\beta(z)|^2-F(z).$$

For $z\neq 0$, completing the square gives $K_c=|q|(H+c)$; the formula for K_c extends smoothly through the regularized collision $z=0$. Here $F=|q|(\Omega-c)$ as in Remark 2.1, and the bounded component is taken away from the distant collision $q=e$. The rescaling $w=4v$ is a fixed invertible linear change of the ambient coordinates and preserves strict convexity. On $K_c=0$,

write $\eta = v + \beta(z) = \sqrt{F}(\cos\phi, \sin\phi)$. The ambient affine coordinates remain (z, v); η is only a fiber parametrization, not a nonlinear change of the embedding used to test convexity. Direct differentiation gives $\operatorname{Hess}\beta_1 = M_1$ and $\operatorname{Hess}\beta_2 = M_2$, with the matrices below.

We use the following scalar-gate formulation of the LC convexity check. Its data are encoded by the symmetric 2×2 matrix pencil

$$L(z, \phi) = -\tfrac{1}{2}\operatorname{Hess} F(z) + \sqrt{F}\,\big(\cos\phi\, M_1(z) + \sin\phi\, M_2(z)\big), \tag{2}$$

where $\phi \in \mathbb{S}^1$ is the fiber angle and M_1, M_2 are the fixed structure matrices arising from the second-order data of the LC map,

$$M_1 = \begin{pmatrix} -4z_2 & -4z_1 \\ -4z_1 & -12z_2 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 12z_1 & 4z_2 \\ 4z_2 & 4z_1 \end{pmatrix}. \tag{3}$$

The two scalar *gates* include both the gradient and the matrix data:

$$T(z, \phi) := |F_z|^2 + 4F\ \operatorname{tr} L(z, \phi), \tag{4}$$

$$D(z, \phi) := F_z^{\top}\operatorname{adj}\big(L(z, \phi)\big)F_z + 4F\ \det L(z, \phi), \tag{5}$$

where $F_z = (\partial_{z_1} F, \partial_{z_2} F)^{\top}$ is the base gradient and adj denotes the adjugate. For $F > 0$, define

$$C(z, \phi) := L(z, \phi) + \frac{F_z F_z^{\top}}{4F}.$$

The rank-one determinant identity gives $T = 4F \operatorname{tr} C$ and $D = 4F \det C$. Thus $T, D > 0$ is equivalent to positive-definiteness of C, not necessarily of L. At $F = 0$ we use the original, nonsingular gate formulas (4)–(5). For completeness, the scalar criterion follows directly from the second fundamental form. For a variation $(\xi, \delta v)$ set $u = \delta v + D\beta(z)\xi$ and $g = F_z$. The tangent constraint and half of the Hessian quadratic form are

$$2\eta \cdot u = g \cdot \xi, \qquad \frac{1}{2}\operatorname{Hess} K_c[(\xi, \delta v), (\xi, \delta v)] = |u|^2 + \xi^{\top} L \xi.$$

On a regular level the second fundamental form, with the sign convention positive on the outward-oriented sphere, is $\operatorname{Hess} K_c/|\nabla K_c|$ restricted to the tangent space, so its definiteness is that of the displayed quadratic form. If $F > 0$, put $e_\phi = \eta/\sqrt{F}$ and write

$$u = \frac{g \cdot \xi}{2\sqrt{F}} e_\phi + tJe_\phi, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The restricted quadratic form is therefore $t^2 + \xi^{\top} C \xi$. It is positive definite if and only if C is positive definite, equivalently $T > 0$ and $D > 0$ by the trace and determinant identities above.

At a collapsed fiber $F = 0$, one has $\eta = 0$, $L = L_0 = -\frac{1}{2}\operatorname{Hess} F$, and the tangent constraint is $g \cdot \xi = 0$. Regularity is equivalent to $g \neq 0$ there. Writing $\xi = \lambda Jg/|g|$ gives

$$|u|^2 + \frac{\lambda^2}{|g|^2}(Jg)^{\top} L_0 (Jg) = |u|^2 + \frac{\lambda^2 D}{|g|^2}, \qquad T = |g|^2,$$

where $(Jg)^{\top} L_0 (Jg) = g^{\top} \operatorname{adj}(L_0) g = D$. Thus the nonsingular gate formulas also give precisely the positive second fundamental form and regularity at $F = 0$. The formulas for K_c, F, and β are smooth at $z = 0$, where $F = 1 - \mu > 0$, so the interior argument includes the regularized collision as well. This establishes the gate criterion on the entire regularized component; it does not assert that positivity has been numerically validated for a particular parameter pair or parameter box.

3. Exact fiber elimination

Throughout this section μ, c are fixed and z ranges over $B_{\mu,c}$, where $F \geq 0$ (the closed component bounded by the collar), so that $\sqrt{F}$ is real.

3.1. **The tangential gate.**

Lemma 3.1 (Closed form for the tangential gate)**.** *With T as in (4) and L as in (2),*

$$T(z,\phi) = \mathcal{T}(z) + 64\,F^{3/2}\big(\sqrt{s} - z_2\cos\phi + z_1\sin\phi\big), \qquad \mathcal{T}(z) := |F_z|^2 - 2F\,\Delta F - 64\,F^{3/2}\sqrt{s}, \tag{6}$$

where $\Delta F = \operatorname{tr}\operatorname{Hess} F$. The bracket in (6) is nonnegative, and

$$\min_{\phi\in\mathbb{S}^1} T(z,\phi) = \mathcal{T}(z), \tag{7}$$

attained, for $s>0$, at $(\cos\phi^*,\sin\phi^*) = (z_2,-z_1)/\sqrt{s}$. If $s=0$, then $z=0$ and the angular term has zero amplitude, so every angle is minimizing. Every angle is also minimizing on the collar $F=0$. Consequently the tangential gate holds on the whole fiber, $T(z,\phi)>0$ for all ϕ, if and only if $\mathcal{T}(z)>0$.

Proof. From (3), $\operatorname{tr} M_1 = -4z_2 - 12z_2 = -16z_2$ and $\operatorname{tr} M_2 = 12z_1 + 4z_1 = 16z_1$. Hence, by (2),

$$\operatorname{tr} L = -\tfrac{1}{2}\Delta F + \sqrt{F}\,(-16z_2\cos\phi + 16z_1\sin\phi),$$

and substituting into (4),

$$T = |F_z|^2 - 2F\,\Delta F + 64\,F^{3/2}(-z_2\cos\phi + z_1\sin\phi).$$

Adding and subtracting $64F^{3/2}\sqrt{s}$ gives (6). The map $\phi \mapsto -z_2\cos\phi + z_1\sin\phi$ is a sinusoid of amplitude $\sqrt{z_1^2+z_2^2} = \sqrt{s}$, so its minimum over $\mathbb{S}^1$ is $-\sqrt{s}$, whence the bracket $\sqrt{s} - z_2\cos\phi + z_1\sin\phi$ ranges over $[0, 2\sqrt{s}]$ and is nonnegative; its minimum 0 is attained when $-z_2\cos\phi + z_1\sin\phi = -\sqrt{s}$; for $s>0$, this is attained at $(\cos\phi^*,\sin\phi^*) = (z_2,-z_1)/\sqrt{s}$. Since $F\ge 0$, the term $64F^{3/2}(\cdot)$ is nonnegative and vanishes at this angle. If $s=0$ or $F=0$, the angular contribution is identically zero. This proves (7) in all cases. The amplitude claim is the exact identity

$$s - (-z_2\cos\phi + z_1\sin\phi)^2 = (z_1\cos\phi + z_2\sin\phi)^2, \tag{8}$$

a perfect square, which shows $(-z_2\cos\phi + z_1\sin\phi)^2 \le s$ for every ϕ. □

The point of Lemma 3.1 is that (7) is an *exact minimum*, not a bound: the tangential gate is certified over the whole fiber precisely when the two-variable surrogate $\mathcal{T}$ is positive. No fiber sweep is required.

Lemma 3.2 (Base-invariant reduction)**.** *The quantities $|F_z|^2$ and ΔF are functions of (s,a) alone; explicitly,*

$$|F_z|^2 = 4s\,F_s^2 + 8a\,F_sF_a + 4s\,F_a^2, \tag{9}$$

$$\Delta F = 4s\,F_{ss} + 8a\,F_{sa} + 4s\,F_{aa} + 4F_s, \tag{10}$$

where subscripts denote partial derivatives of F regarded as a function of (s,a). In particular $\mathcal{T} = \mathcal{T}(s,a;c)$.

Proof. Since F depends on z through $s = z_1^2+z_2^2$ and $a = z_1^2 - z_2^2$, the chain rule gives $\partial_{z_1}F = 2z_1(F_s + F_a)$ and $\partial_{z_2}F = 2z_2(F_s - F_a)$. Therefore

$$|F_z|^2 = 4z_1^2(F_s+F_a)^2 + 4z_2^2(F_s-F_a)^2 = 4(z_1^2+z_2^2)(F_s^2+F_a^2) + 8(z_1^2-z_2^2)F_sF_a,$$

which is (9). A second application of the chain rule yields $\partial_{z_1}^2F + \partial_{z_2}^2F = 4sF_{ss} + 8aF_{sa} + 4sF_{aa} + 4F_s$, which is (10). Both are functions of (s,a), and by Lemma 3.1 so is $\mathcal{T}$. □

Corollary 3.3 (Quadrant reduction)**.** *The reduced tangential gate $\mathcal{T}$ is invariant under the reflections $(z_1,z_2)\mapsto(\pm z_1,\pm z_2)$. Hence the tangential gate need only be certified on a single closed quadrant of the base.*

Proof. Both $s = z_1^2+z_2^2$ and $a = z_1^2 - z_2^2$ are invariant under each sign change of z_1 or z_2, and $\mathcal{T}$ depends on z only through (s,a) by Lemma 3.2. □

Corollary 3.3 supplies a rigorous justification for the quadrant symmetry reduction used in practice: it is not an assumption but a consequence of the functional form of F.

3.2. **The curvature-determinant gate.** The elimination of ϕ from D rests on a purely multilinear identity, valid for arbitrary symmetric 2×2 data, which we state and prove in that generality before specializing.

For symmetric 2×2 matrices X, Y write $\sigma(X,Y) := \operatorname{tr}X\,\operatorname{tr}Y - \operatorname{tr}(XY)$ for the polarization of the determinant, so that $\det(X+Y) = \det X + \det Y + \sigma(X,Y)$; recall that for 2×2 matrices adj is linear.

Lemma 3.4 (Harmonic decomposition of the determinant gate)**.** *Let L_0, M_1, M_2 be symmetric 2×2 matrices, $g\in\mathbb{R}^2$, $\rho\ge 0$, and $F\in\mathbb{R}$. Put $v = (\rho\cos\phi, \rho\sin\phi)$ and $L = L_0 + v_1M_1 + v_2M_2$. Then*

$$g^{\top}\operatorname{adj}(L)\,g + 4F\det L = A_0 + A_1\cos\phi + B_1\sin\phi + A_2\cos 2\phi + B_2\sin 2\phi, \tag{11}$$

with

$$A_0 = g^{\top}\operatorname{adj}(L_0)g + 4F\det L_0 + 2F\rho^2\big(\det M_1 + \det M_2\big),$$
$$A_1 = \rho\big(g^{\top}\operatorname{adj}(M_1)g + 4F\,\sigma(L_0,M_1)\big), \qquad B_1 = \rho\big(g^{\top}\operatorname{adj}(M_2)g + 4F\,\sigma(L_0,M_2)\big),$$
$$A_2 = 2F\rho^2\big(\det M_1 - \det M_2\big), \qquad B_2 = 2F\rho^2\,\sigma(M_1,M_2).$$

Proof. By linearity of the adjugate, $\operatorname{adj}(L) = \operatorname{adj}(L_0) + v_1\operatorname{adj}(M_1) + v_2\operatorname{adj}(M_2)$, so

$$g^{\top}\operatorname{adj}(L)g = g^{\top}\operatorname{adj}(L_0)g + v_1\,g^{\top}\operatorname{adj}(M_1)g + v_2\,g^{\top}\operatorname{adj}(M_2)g.$$

By the polarization identity for 2×2 determinants applied twice,

$$\det L = \det L_0 + v_1^2\det M_1 + v_2^2\det M_2 + v_1\sigma(L_0,M_1) + v_2\sigma(L_0,M_2) + v_1v_2\,\sigma(M_1,M_2).$$

Substitute $v_1 = \rho\cos\phi$, $v_2 = \rho\sin\phi$ and use the double-angle identities $\cos^2\phi = \frac{1}{2}(1+\cos 2\phi)$, $\sin^2\phi = \frac{1}{2}(1-\cos 2\phi)$, $\cos\phi\sin\phi = \frac{1}{2}\sin 2\phi$. Collecting the constant, first-harmonic and second-harmonic terms gives exactly the stated coefficients. □

Specializing to the LC surface, take $L_0 = -\frac{1}{2}\operatorname{Hess}F$, $g = F_z$, and $\rho = \sqrt{F}$ (so that $\rho^2 = F$), with M_1, M_2 as in (3). Then (11) is precisely the fiber expansion of the gate D in (5). The next lemma computes the second-harmonic amplitude for these structure matrices.

Lemma 3.5 (Second-harmonic amplitude)**.** *For M_1, M_2 as in (3),*

$$\det M_1 + \det M_2 = 32s, \qquad (\det M_1 - \det M_2)^2 + \sigma(M_1,M_2)^2 = (64s)^2. \tag{12}$$

Consequently, with $\rho^2 = F$, the second-harmonic amplitude of D is

$$\sqrt{A_2^2 + B_2^2} = 2F\rho^2\sqrt{(\det M_1 - \det M_2)^2 + \sigma(M_1,M_2)^2} = 128\,F^2 s. \tag{13}$$

Proof. Direct computation from (3) gives $\det M_1 = 48z_2^2 - 16z_1^2$ and $\det M_2 = 48z_1^2 - 16z_2^2$, whence $\det M_1 + \det M_2 = 32(z_1^2+z_2^2) = 32s$ and $\det M_1 - \det M_2 = -64(z_1^2 - z_2^2) = -64a$. A short calculation gives $\sigma(M_1,M_2) = -128\,z_1z_2$, so

$$(\det M_1 - \det M_2)^2 + \sigma(M_1,M_2)^2 = 4096a^2 + 16384z_1^2z_2^2 = 4096(a^2 + 4z_1^2z_2^2) = 4096\,s^2 = (64s)^2,$$

using $a^2 + 4z_1^2z_2^2 = (z_1^2 - z_2^2)^2 + 4z_1^2z_2^2 = (z_1^2+z_2^2)^2 = s^2$. Substituting into $\sqrt{A_2^2+B_2^2} = 2F\rho^2\sqrt{(\det M_1 - \det M_2)^2 + \sigma(M_1,M_2)^2}$ with $\rho^2 = F$ gives (13). □

Corollary 3.6 (Rigorous lower bound for the determinant gate)**.** *For every $\phi\in\mathbb{S}^1$,*

$$D(z,\phi) \ge \mathcal{D}(z) := A_0(z) - \sqrt{A_1(z)^2 + B_1(z)^2} - 128\,F(z)^2 s, \tag{14}$$

where A_0, A_1, B_1 are as in Lemma 3.4 with the specialization above. The bound is sharp on the collar: at $\{F=0\}$ one has $A_1 = B_1 = A_2 = B_2 = 0$ and $D = \mathcal{D} = F_z^{\top}\operatorname{adj}(-\frac{1}{2}\operatorname{Hess}F)F_z$.

Proof. By Lemma 3.4 and the elementary bounds $|A_1\cos\phi+B_1\sin\phi| \le \sqrt{A_1^2+B_1^2}$ and $|A_2\cos 2\phi+B_2\sin 2\phi| \le \sqrt{A_2^2+B_2^2} = 128F^2 s$ (the last equality by Lemma 3.5), we get $D \ge A_0 - \sqrt{A_1^2+B_1^2} - 128F^2 s$. On the collar $F=0$: the coefficients A_1, B_1 carry a factor $\rho=\sqrt{F}$ and A_2, B_2 a factor $\rho^2 = F$, so all four vanish, leaving $D = A_0 = F_z^{\top}\,\mathrm{adj}(-\tfrac{1}{2}\,\mathrm{Hess}\,F)F_z = \mathcal{D}$. □

3.3. **The reduction theorem.** For clarity, write

$$D_{\min}(z) := \min_{\phi\in\mathbb{S}^1} D(z,\phi).$$

Here $D(z,\phi)$ is the determinant gate, $\mathcal{D}(z)$ is its rigorous amplitude lower bound, and $D_{\min}(z)$ is its exact fiber minimum. They satisfy $\mathcal{D}(z) \le D_{\min}(z)$, with equality on the collar; away from the collar equality need not hold. Combining the lemmas therefore gives an exact tangential test and a sufficient determinant test in two base variables.

Theorem 3.7 (Two-dimensional sufficient criterion for the convexity gates)**.** *Fix μ, c and let z range over $B_{\mu,c}$. Then:*

(a) *The tangential gate holds on every fiber, $T(z,\phi) > 0$ for all $\phi \in \mathbb{S}^1$, if and only if $\mathcal{T}(z) > 0$, where $\mathcal{T} = |F_z|^2 - 2F\Delta F - 64F^{3/2}\sqrt{s}$ is a function of (s,a) alone.*

(b) *The determinant gate holds on every fiber whenever $\mathcal{D}(z) > 0$, where $\mathcal{D} = A_0 - \sqrt{A_1^2+B_1^2} - 128F^2 s$; the sufficient condition $\mathcal{D} > 0$ is sharp on the collar, where it coincides with D.*

In particular, strict convexity of $\Sigma_{\mu,c}$ is certified by the two-variable conditions $\mathcal{T} > 0$ and $\mathcal{D} > 0$ over the base, with no sweep over the fiber angle.

Proof. Part (a) is Lemmas 3.1 and 3.2. Part (b) is Corollary 3.6. The final statement combines the two, since positivity of both gates over (z,ϕ) certifies strict convexity by the criterion of Section 2, and $\mathcal{T} > 0$ is equivalent to $\min_\phi T > 0$ while $\mathcal{D} > 0$ implies $\min_\phi D > 0$. □

Remark 3.8 (Symbolic certification)**.** Each identity above—the gap formula (6), the perfect square (8), the attained angle for $s > 0$, the (s,a) reductions (9)–(10), the generic harmonic decomposition (11), and the amplitude (12)—was verified by an exact computer-algebra audit. The audit forms each residual as the difference of the two sides and simplifies it to the symbol 0; all eight residuals vanish identically. The script is deterministic, runs in seconds, and is reproduced in Appendix A.

Remark 3.9 (Effect on certification cost)**.** The elimination removes the innermost loop of the certificate. In an interval-arithmetic setting, the fiber sweep is replaced by the evaluation of two explicit two-variable functions, and the trigonometric enclosure inflation of $\cos\phi, \sin\phi$ over $[0, 2\pi)$ is eliminated in the tangential gate (where the minimum is exact) and reduced to a single amplitude bound in the determinant gate.

4. Application: an Earth–Moon numerical spot check

As a finite numerical spot check, we evaluate the reduced gates on a small rectangle near the Earth–Moon mass ratio (by Remark 2.1, with $\mu \approx 0.9878$ the regularized component is the *Moon's*, $m_r \approx 0.0122$),

$$\mu \in [0.987839414390376,\ 0.987859414390376], \qquad c \in [1.5995,\ 1.6005].$$

At the four parameter corners, using 400 first-quadrant rays and 160 radial rows, including the numerically located collar, the minimum sampled value of $\mathcal{T}$ is at least 2.29×10^{-3} and the minimum sampled collar value of $\mathcal{D}$ is at least 6.95×10^{-4}. These positive sampled values are consistent with $\mathcal{T} = \min_\phi T$ and with the collar identity $\mathcal{D} = D_{\min}$ at $F = 0$. Collar evaluations use the nonsingular $F = 0$ formulas at the computed radial roots; these are floating-point evaluations, not validated root enclosures.

This paragraph is deliberately only a numerical spot check. It does not provide an interval certificate on the full parameter rectangle, does not validate all base and fiber points, and is not used as a premise for the exact fiber-elimination identities of Section 3. A full certificate on a parameter rectangle would require validated evaluation of the reduced functions over a complete subdivision of that rectangle and its base domains.

5. Global convexity reconnaissance

We now use the reduced gates for a floating-point reconnaissance of LC-convexity across a selected mass–energy grid. This section is exploratory; its computed results are floating-point evidence and are not interval-certified. In particular, numerical values reported for $D_{\min}$ are sampled approximations to the exact minimum defined in Section 3, not validated enclosures.

5.1. **Method.** For a grid of mass ratios and subcritical energies we compute the reduced tangential surrogate $\mathcal{T}$ and, for the determinant gate, a sampled approximation to $D_{\min}$ by evaluating the exact degree-two trigonometric polynomial of Lemma 3.4 on a dense angular sample over the union of the worst base points of the amplitude bound $\mathcal{D}$ *and* the entire collar band (the outermost radial rows on every ray; the bound is exact at $F = 0$, but may be conservative at nearby interior points). The map uses 192 first-quadrant base rays, 80 radial rows, and 256 equally spaced fiber angles; the selected base set is the union of the 80 worst amplitude-bound points and the outermost three radial rows on every ray. At the numerically located collar, the nonsingular $F = 0$ formulas are evaluated explicitly, avoiding square roots of small negative root residuals. This boundary treatment is a numerical convention, not a validated enclosure of the roots. The harmonic formula is exact, but minimizing it over a finite angular sample is not exact minimization over the circle. Positive sampled values alone do not certify positivity over the full fiber or base. The collar band is essential: because $\mathcal{D}$ is loose in the interior, its worst points can all be interior while the true minimum of D sits on the collar. The first critical energy $c_{L_1}(\mu)$ is located by robust bisection between the primaries; as a sanity check the equal-mass value is recovered as $c_{L_1} = 2.000000$. The grid spans sixteen mass values (including both Earth–Moon components, $\mu = 0.987849\ldots$ and $\mu = 0.012151\ldots$) and seven energy offsets $\Delta c = c - c_{L_1}$; the validation script `audit/v130_validation.py` cross-checks nine selected sign conditions on a specified finite base×fiber grid, without the worst-point heuristic; it does not validate every numerical value quoted in the paper.

5.2. **The non-convexity window.** The computed tangential surrogate $\mathcal{T}$ is positive at the sampled points. The determinant gate, by contrast, has negative sampled values in a large, asymmetric near-critical window: writing $m_r = 1 - \mu$ for the regularized-primary mass fraction, at $\Delta c = 10^{-4}$ the computed approximation to $D_{\min}$ is negative for every grid value $m_r \geq 0.059$ up to and including $m_r = 0.999$, and positive for $m_r \leq 0.03$; dedicated finite scans at this offset have positive sampled gates at $m_r = 0.04$ and a negative sampled determinant at $m_r = 0.05$, bracketing the sampled sign transition. On the 600-ray, 400-row, 1024-fiber-angle grid, the sampled determinant minima are approximately $+3.89 \times 10^{-5}$ and -5.75×10^{-6}, respectively; their magnitudes depend on resolution. At this sampled offset, the scans suggest LC-convexity only for components regularized at a sufficiently light primary; this is not a certified classification. The window shrinks from the light side as the energy recedes from critical and is empty for $\Delta c \gtrsim 0.03$ at every mass ratio scanned; the heavy side persists longest (at $\Delta c = 10^{-2}$ the negative grid entries are $m_r \in \{0.9, 0.95, 0.98, 0.987849\ldots\}$, including the Earth component). On the heavy side the lobe depth decreases toward the rotating-Kepler limit ($D_{\min} \approx -0.58$ at $m_r = 0.999$, $\Delta c = 10^{-4}$, in the dedicated 600-ray, 240-row, 512-fiber-angle check), but these data do not establish a limiting value as $m_r \to 1$. At the sampled points the observed failure is of the determinant gate, while the tangential gate stays positive. Figure 1 shows the resulting phase map.

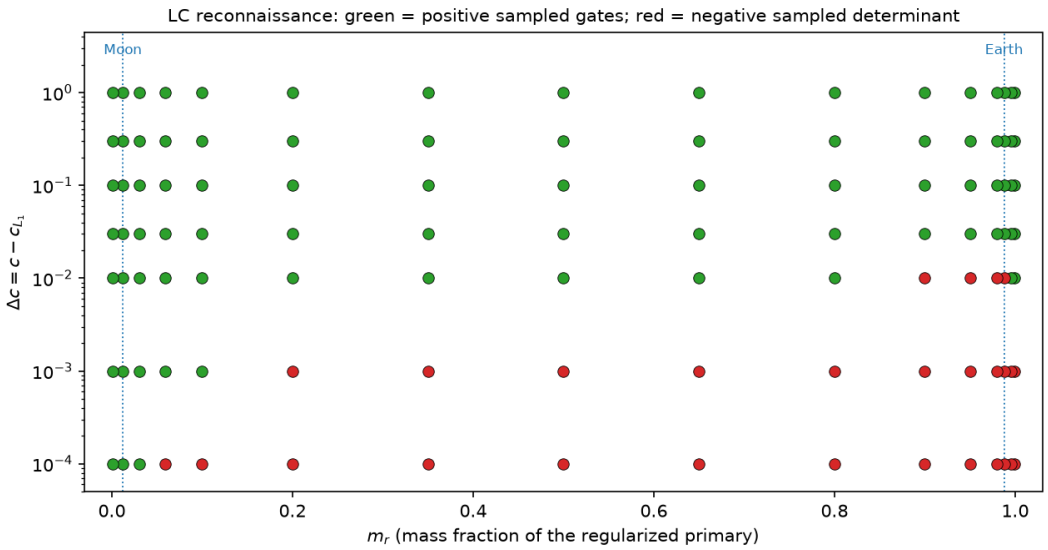


FIGURE 1. Floating-point reconnaissance of LC-convexity in the regularized-primary fraction m_r (Remark 2.1). Green markers: no negative gate value detected by the sampling procedure. Red markers: a sampled determinant value is negative. The Earth and Moon fractions are indicated. These classifications are not interval certificates. No threshold from the Euler problem is asserted for this rotating-model plot; see Section 5.3. The Moon component's smaller near-critical offsets are examined in Section 5.5.

5.3. **Relation to Kim's Euler-problem result.** Kim's LC non-convexity result [5] concerns the Euler problem of two fixed centers and uses the distant-primary fraction μ_K: its condition $\mu_K < 16/17$ corresponds to regularized-primary fraction $1 - \mu_K > 1/17$, not $m_r < 16/17$. Moreover, the Euler Hamiltonian lacks the PCR3BP rotation term. Consequently, this threshold cannot validate the present rotating-model mass–energy map. The sampled sign transition between $m_r = 0.04$ and 0.05 at $\Delta c = 10^{-4}$ and the values through $m_r = 0.999$ are numerical findings for (1), with no theorem-based validation inferred from Kim's threshold.

5.4. **The Earth–Moon system.** The two components behave very differently. The Moon component ($m_r = 0.0122$) lies below the observed light-side boundary and has positive sampled gate values at every energy offset scanned in the map ($\Delta c \geq 10^{-4}$); this is the component used in the numerical spot check of Section 4, at offsets around $\Delta c \approx 5.7 \times 10^{-3}$. The Earth component ($m_r = 0.9878$), by contrast, sits inside the heavy-side window: its sampled determinant is negative at $\Delta c = 10^{-4}$ with a macroscopic margin ($D_{\min} \approx -1.81$), and a dedicated numerical energy sweep places the observed sign transition at $\Delta c^{\mathrm{E}} \in (1.1 \times 10^{-2}, 1.2 \times 10^{-2})$ ($D_{\min} \approx -9.5 \times 10^{-3}$ at $\Delta c = 1.1 \times 10^{-2}$ and $+1.1 \times 10^{-1}$ at 1.2×10^{-2}). The near-critical vicinity of the Moon component is refined next.

5.5. **Refinement at the critical limit: an ultrathin non-convex shell.** The map above has a smallest energy offset of $\Delta c = 10^{-4}$. Refining the scan for the *Moon* component ($m_r = 0.012150585609624$, distant-primary parameter $\mu = 0.987849414390376$, $c_{L_1} = 1.594170558875$) down to $\Delta c = 10^{-7}$, with a log-spaced energy grid, angular resolution up to 600 base rays, 400 radial samples, and the fiber minimum of D evaluated on 1024 angular samples of the exact degree-two polynomial of Lemma 3.4, gives numerical evidence for a feature invisible at the coarser resolution: the sampled determinant minima suggest an ultrathin non-convex shell immediately above the

critical energy,

$$c \in \left(c_{L_1},\, c_{L_1} + \Delta c^*\right], \qquad \Delta c^* \approx 9.54 \times 10^{-6},$$

with the approximate endpoint located by numerical bisection in Δc. Three computational observations characterize the apparent shell. First, the negative values appear robust under refinement: as the sampled offsets Δc decrease toward zero, the computed minima appear to approach a negative limit ($D_{\min} \approx -1.5 \times 10^{-6}$ at $\Delta c = 10^{-6}$ and $\approx -1.7 \times 10^{-6}$ at $\Delta c = 10^{-7}$, the smallest offset probed), and the computed values appear stable under refinement of the radial sampling ($n_s = 60 \to 400$) and of the fiber sampling ($512 \to 1024$ angles). Second, at the sampled points the failure is again of the determinant gate only, with $\mathcal{T} > 0$, consistent with a single negative direction rather than double concavity. Third, the minimizing base point sits at polar angle $\theta^* \approx 0.076$—near the $z_2 = 0$ base axis, without identifying that ray with an orbit—at $s/s^* \approx 0.99$, where s^* is the squared boundary radius on that ray, far from the $s = \frac{1}{2}$ singularity guard of the scan. These refinement checks provide evidence of robustness, but do not rule out numerical error or prove a limiting value.

At $c = c_{L_1}$ the first Lagrange point lies on the critical level, where the bounded Hill component reaches the neck. This provides geometric context for the observed near-critical behavior, not a proof of the shell or a transfer of Kim's Euler-model theorem. The quantitative observation here is an apparent shell of width $\sim 10^{-5}$ in c for the *Moon* component, rather than the macroscopic window observed for the Earth component. For the latter, negative determinant values are found at sampled offsets $\Delta c \lesssim 1.2 \times 10^{-2}$.

Two consequences follow, both stated as numerical observations, and both now component-resolved. For the certification program, the sampled values suggest that the LC-convexity route may apply to the Moon component at offsets $c - c_{L_1} \geq 10^{-5}$; they do not certify this whole energy range. The apparent ultrathin shell $\left(c_{L_1}, c_{L_1} + 10^{-5}\right]$ and, more consequentially, the Earth component's macroscopic window $\left(c_{L_1},\, c_{L_1} + 1.2 \times 10^{-2}\right)$ motivate alternatives to strict convexity in this particular LC embedding, rather than establishing a dynamical conclusion from these numerical scans alone. The numerical spot check in Section 4 concerns a parameter rectangle well separated from this numerically observed shell. Those pointwise checks do not provide or replace an interval certificate.

6. A dynamical-relevance observation

The non-convexity of Section 5 is a property of the LC embedding, not directly of the dynamics. We record here a numerical observation—exploratory, floating-point—bearing on whether the non-convexity is dynamically relevant.

Since F depends only on (s, a), its bounded base is invariant under each coordinate reflection. The matrices in (3) give

$$D(-z_1, z_2, \phi) = D(z_1, z_2, -\phi), \qquad D(z_1, -z_2, \phi) = D(z_1, z_2, \pi - \phi).$$

Thus $D_{\min}$ is even in each base coordinate. Figure 2 uses the first radial zero of F on each actual polar ray and displays the reflection-related sampled regions in all four quadrants.

Observation 6.1. Across the comparable-mass window (distant-primary parameter μ from 0.2 to 0.6, i.e. regularized fractions m_r from 0.8 down to 0.4; $c = c_{L_1} + 10^{-3}$), the sampled base points with $D_{\min} < 0$ form reflection-related lateral lobes hugging the collar: in first-quadrant base polar coordinates, the sampled lobes occupy approximately the radial fraction $s/s^* \in [0.95, 1.0]$ and approximately the angular band $\theta \in [3°, 12°]$, displaced from both coordinate axes by at least 2°–4°. Along the two coordinate axes, the sampled approximation to $D_{\min}$ is positive; for equal masses at $c = c_{L_1} + 10^{-3}$ it is $+2.055 \times 10^{-1}$ along $z_2 = 0$ and $+5.190$ along $z_1 = 0$.

Figure 2 displays the lobes and determinant profiles along selected base rays. These rays are geometric slices, not integrated periodic trajectories. The axial collision-orbit description in Kim's

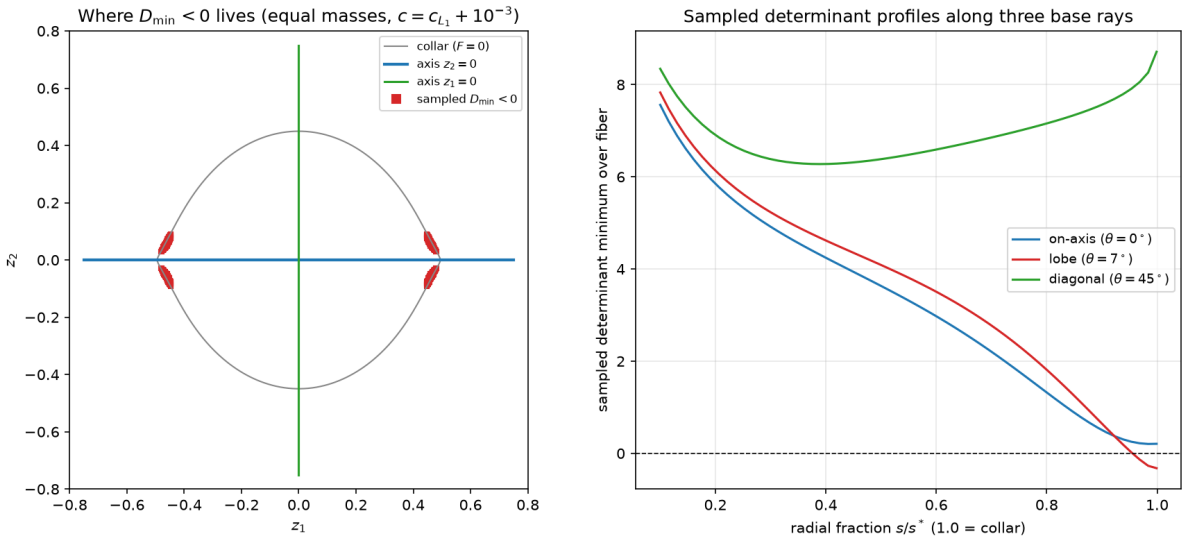


FIGURE 2. Left: sampled negative determinant values near the base collar, with the coordinate axes $z_2 = 0$ and $z_1 = 0$. Right: sampled approximations to $D_{\min}$ along three selected rays. These are base-slice profiles, not computed retrograde or direct trajectories. Floating-point, equal masses, $c = c_{L_1} + 10^{-3}$. The left panel samples 720 full-circle rays and 60 radial fractions in $[0.5, 1]$; each fiber uses 192 angles. The boundary is the first radial zero of F on each ray.

Remark 1.2 in the cited arXiv version belongs to the Euler model; it does not identify the PCR3BP retrograde orbit with either LC coordinate axis. Accordingly, Observation 6.1 does not establish that the retrograde or direct orbit avoids the sampled non-convexity lobes.

We stress the limits of this observation. Dynamical convexity requires an index condition on all contractible closed Reeb orbits, and the rational-section criterion also requires the topology of a proposed binding [4]. A positive gate profile on a base axis supplies neither. The sampled lobes identify locations where this embedding-based test detects a problem, but they do not prove dynamical inertness or exclude orbit visits. Nor should the comparable-mass regime be described as wholly unresolved: the near-equal-mass subcritical conclusion is established in [12, Thm. 1.16].

6.1. **Fine structure of the lobes: one negative direction.** A closer look at the lobes records two further numerical regularities, both observed at sampled masses in the comparable-mass band $\mu \in [0.2, 0.8]$ (equivalently $m_r \in [0.2, 0.8]$) at $c = c_{L_1} + 10^{-3}$. They sharpen Observation 6.1 from "$D_{\min}$ has negative sampled values off-axis" to a statement about *how* it becomes negative.

Observation 6.2. On the grid of 200 first-quadrant rays, 80 radial fractions in $[0.5, 0.9999]$, and 192 fiber angles, the reduced tangential gate $\mathcal{T}$ remains positive at the sampled lobe points: the endpoint sampled minima of $\mathcal{T}$ over $\{D_{\min} < 0\}$ are $+7.5 \times 10^{-1}$ at $\mu = 0.20$ and $+2.6 \times 10^{-1}$ at $\mu = 0.80$, while the corresponding endpoint minima of $D_{\min}$ are -1.69 and -1.4×10^{-3}. No monotonicity of these sampled extrema in the mass parameter is asserted. The observed signs are consistent with *one negative direction*: a negative determinant together with a positive trace corresponds to exactly one negative direction, rather than double concavity. The numerical scans do not certify these signs at every point of the lobes.

For $F > 0$, the matrix C defined in Section 2.2 satisfies $T = 4F \operatorname{tr} C$ and $D = 4F \det C$. Thus at a fiber direction with $D < 0 < \mathcal{T} \le T$, the symmetric matrix C has one positive and one negative eigenvalue. The scans neither certify all points of the lobes nor determine the dynamics there. Figure 3 displays the sampled gate profiles.

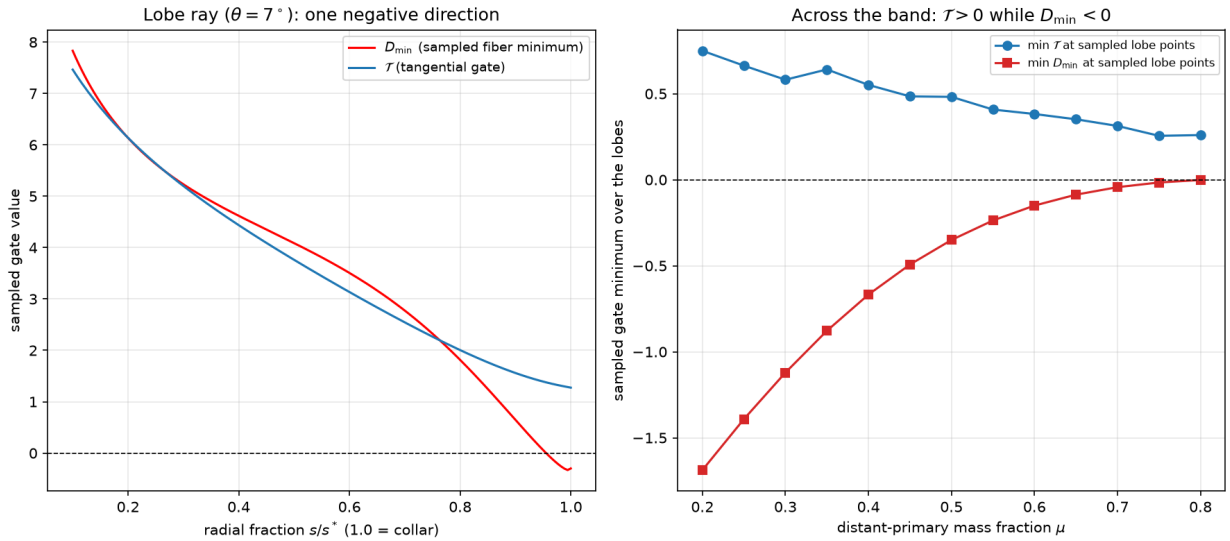


FIGURE 3. Fine structure of the lobes (floating-point, $c = c_{L_1} + 10^{-3}$). Left: along a sampled lobe ray ($\theta = 7^\circ$, equal masses), the approximation to $D_{\min}$ dips below zero near the collar while the sampled $\mathcal{T}$ stays positive. Right: over the sampled mass values and lobe points, the computed minimum of $\mathcal{T}$ is positive and that of $D_{\min}$ is negative. The plotted $D_{\min}$ values are sampled fiber minima, not the rigorous bound $\mathcal{D}$. The right panel uses $\mu = 0.20, 0.25, \ldots, 0.80$ with the grid of Observation 6.2; the left panel uses 160 radial fractions and 192 fiber angles.

A companion regularity concerns the radial geometry rather than the fiber. On the sampled rays the computed collar derivative satisfies $\partial F/\partial r < 0$ (the collar radial derivative ranges from -2.5×10^{-1} at $\mu = 0.20$ to -1.0×10^{-1} at $\mu = 0.80$), on rays through the lobes no less than on-axis. These values support radial star-shapedness of the bounded base and suggest that the sampled non-convexity lobes do not erode this radial-derivative margin, without identifying radial star-shapedness of this two-dimensional base with contact type of the full regularized hypersurface or with the hypotheses of a global-section theorem [1, 4]. Both regularities are reproduced by `code/lobe_structure.py`; the complete data and Figures 2–3 are generated by `code/audit_figures.py` in the reproducibility package; like Observation 6.1 they are floating-point at a finite grid and are stated as such.

7. LIMITATIONS AND OUTLOOK

The identities of Section 3 give an exact tangential minimum and a sufficient determinant lower bound. Where both reduced functions are positive, they remove the fiber sweep from the LC-convexity test. The numerical reconnaissance of Section 5 suggests favorable sampled regimes for light-primary components (positive sampled gates at $m_r = 0.04$ and for the Moon component of the Earth–Moon pair at $\Delta c = 10^{-4}$, with a negative sampled determinant at $m_r = 0.05$), and at energy offsets $\Delta c \gtrsim 0.03$ for the mass values scanned (for the Earth component, $\Delta c \gtrsim 1.2 \times 10^{-2}$). Two limitations bound its reach.

First, the reduced determinant gate uses the sufficient condition $\mathcal{D} > 0$, which is sharp only on the collar; away from the collar $\mathcal{D}$ may be conservative, and a base point with $\mathcal{D} < 0 < D_{\min}$ would require the full harmonic minimum of Lemma 3.4 rather than the amplitude bound. This is a matter of tightness, not of correctness.

Second, failure of this LC embedding to be convex must be separated from failure of dynamical convexity or of a global section. The scans suggest a broad near-critical obstruction to this particular convexity test, but they do not certify a continuous parameter boundary. Beyond this test, two approaches are relevant: (i) verifying the required Conley–Zehnder and binding conditions, as in [4, 12, 13]; and (ii) finding a suitable symplectic embedding, as in the rotating-Kepler and near-equal-mass results [10, 12]. The near-equal-mass subcritical conclusion of [12] is an external result, not a consequence of the present scans. In the midpoint-centered elliptic coordinates (λ, ν) of [5], the rotation term has the exact identity

$$(\cosh^2\lambda - \cos^2\nu)\,(q_1p_2 - q_2p_1) = \tfrac{1}{2}\big(p_\lambda \sin 2\nu + p_\nu \sinh 2\lambda\big), \tag{15}$$

a closed form linear in the momenta. In this midpoint-centered convention, adding a nonzero multiple of the rotation term introduces linear momentum terms into the separated Euler Hamiltonian. The mechanical-case curvature argument therefore cannot be applied by substitution without further justification. This identity alone is not a proof of convexity or non-convexity for the rotating model; in particular, a change from midpoint-centered to barycentric coordinates must account for the associated translation terms. The fiber identities of Section 3 remain independent of that outlook. The present reconnaissance supplies no additional orbit-index or global-section theorem.

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Data and Code Availability

The manuscript sources, code, numerical data, commands, scientific logs, and generators for Figures 1–3 are available at `https://github.com/cagrisa-creator/pcr3bp-fiber-elimination`. The file `REPRODUCIBILITY.txt` documents the dependencies, sampling grids, and correspondence between results and files; `SHA256SUMS_R07.txt` records the file checksums. The package contains exact symbolic audits, numerical samples and negative witnesses, but no interval convexity certificate.

Appendix A. The symbolic audit

The following identities are checked by forming each residual and simplifying it to zero with a computer-algebra system. All eight residuals vanish identically; the driver prints `PASS_C82_FIBER_ELIMINATION_AUDIT`. The complete script is included in the reproducibility package as `code/fiber_elimination_audit.py`.

- R1 Gap formula (6): $T - \mathcal{T} - 64F^{3/2}(\sqrt{s} - z_2\cos\phi + z_1\sin\phi) = 0$.
- R2 Perfect square (8): $s - (-z_2\cos\phi + z_1\sin\phi)^2 - (z_1\cos\phi + z_2\sin\phi)^2 = 0$.
- R3 Attained angle for $s > 0$: $\sqrt{s} - z_2(z_2/\sqrt{s}) + z_1(-z_1/\sqrt{s}) = 0$. At $s = 0$, the angular term vanishes directly, as noted in Lemma 3.1.
- R4 Gradient reduction (9): $|F_z|^2 - (4sF_s^2 + 8aF_sF_a + 4sF_a^2) = 0$.
- R5 Laplacian reduction (10): $\Delta F - (4sF_{ss} + 8aF_{sa} + 4sF_{aa} + 4F_s) = 0$.
- R6 Generic harmonic decomposition (11): difference of the two sides $= 0$ for arbitrary symmetric L_0, M_1, M_2, vector g, and scalars ρ, F.
- R7 Determinant sum (12): $\det M_1 + \det M_2 - 32s = 0$.
- R8 Amplitude (12): $(\det M_1 - \det M_2)^2 + \sigma(M_1, M_2)^2 - (64s)^2 = 0$.

Separately, the closed form (15) for the regularized rotation term in the doubly-covered elliptic coordinates, quoted in Section 7, is verified both symbolically (residual simplifies to 0) and numerically (maximum deviation $< 10^{-13}$ over 2×10^5 random points) by `code/rotation_term_audit.py`, which prints `PASS_ROTATION_TERM_AUDIT`. The coordinate reconstruction of Section 2.2 and the reflection identities of Section 6 are checked independently by

`audit/criterion_symbolic_audit.py`; these are exact identity checks, not numerical positivity certificates.

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